

SE-Lab1

Name: Aashlesh P

SRN: PES1UG24CS011

Section: 5A

Problem Statement #11 – Telemedicine Slot Booking & Prescription Portal

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR001	Functional	The system shall enable physicians to author and digitally sign structured prescriptions that are automatically compiled into downloadable PDF records	High	Pass: Valid cryptographic signature attached to prescription PDF; Fail: Prescription generated without doctor license validation.	Ensures prescriptions are authentic and securely issued by authorized physicians.
FR002	Functional	The system shall allow patients to view doctor specialties and available consultation slots.	High	Pass: The patient selects a specialty and the system displays the available doctors and their consultation slots. Fail: Available doctors or slots are not displayed.	Allows patients to find a suitable doctor and consultation time.
		The system shall allow patients to		Pass: The selected available slot is successfully booked and confirmation is displayed.	Provides the core functionality

FR003	Functional	book an available video consultation slot.	High	Fail: The slot remains available after successful booking or confirmation is not displayed.	for scheduling telemedicine consultations .
FR004	Functional	The system shall provide an encrypted consultation room link to the patient and attending physician after a successful booking.	High	Pass: A secure consultation room link is generated and provided to both the patient and physician. Fail: A consultation link is not generated or is provided without encryption.	Enables secure remote communication between the patient and physician.
FR005	Functional	The system shall allow patients to download digitally signed prescription PDFs issued by the attending physician.	High	Pass: The patient can download the prescription as a PDF containing the physician's digital signature. Fail: The prescription cannot be downloaded	Allows patients to securely access and retain their prescribed medication records.

				or does not contain the digital signature.	
NFR001	Non-Functional	The system shall encrypt all video session metadata and patient clinical notes in transit using TLS 1.3 and at rest using AES-256.	High	Pass: Security testing confirms TLS 1.3 is used for data in transit and AES-256 is used for stored clinical data. Fail: Either requirement is not satisfied.	Protects sensitive patient and consultation information from unauthorized access.
NFR002	Non-Functional	The system shall confirm a successful consultation-slot booking within 3 seconds under a load of at least 500 concurrent users.	Medium	Pass: At least 95% of successful booking requests receive confirmation within 3 seconds during testing with 500 concurrent users. Fail: Fewer than 95% meet the 3-second requirement.	Ensures the booking system remains responsive during periods of high demand.